

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.
5. Use of Z-Table, T-Table and F-Distribution table is allowed.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

12

- A) What do you mean by Statistical Analysis? Explain in detail PPDAC Cycle. (BT2/CO1) 6
- B) Discuss the Following (BT2/CO1) 6
- i) Sampling and Sampling Size.
 - ii) Computational Statistics.
- C) Explain in detail Types of error, Statistical significance, and Confidence Interval. (BT2/CO1) 6

Q.2 Solve Any Two of the following.

12

- A) Find the Quartiles (Q1, Q2, Q3), Range, IQR, Standard Deviation, Variance and Coefficient of Variation for the below data frame. (BT1/CO2) 6

Year	Item 1	Item 2
2010	13	7
2011	18	8
2012	20	4
2013	23	6
2014	26	9
2015	28	11
2016	30	12

- B) Explain the Following (BT4/CO2) 6
- i) Measure of Central Tendency.
 - ii) Kurtosis along with its types.
- C) Enlist all the Key-Functions and Explain any two functions with its formula. (BT2/CO2) 6

Q. 3 Solve Any Two of the following.

12

- A) Explain in detail Box-Cox and power transforms. Find the Transformed data for given λ values (-3, -2, -1, -0.5, 0, 0.5, 1, 2, 3). (BT2/CO3) 6
- B) Explain Logit transform with its back transform and graph in detail. (BT2/CO3) 6
- C) Discuss the following. (BT2/CO3) 6

- i) The Log and Exponential transforms with Back Transforms.
- ii) Write the Lognormal Distribution, Bounded and Un-Bounded Variation Formula.

Q.4 Solve Any Two of the following. **12**

- A)** What is goodness of fit test? List the types and explain any one test in detail. **(BT2/CO4) 6**
- B)** Solve the Following **(BT3/CO4) 6**
 - i) The city survey taken by Food-Plaza claiming that 13 out of 20 hotels serve the hygienic food with good service. To test that claim random sample of 120 hotels from city is obtained. Out of those 120 hotels, 89 indicate that they serve the hygienic food with good service. Is this claim accurate? Use $\alpha = 0.05$. (Calculate Result using Appropriate Test)
 - ii) Apply T-Test for the data set having population mean 14
Data Set : (11, 12, 12, 13, 13, 14, 14, 15, 15, 16)
- C)** What do you mean by Variance test? Explain the F-test for two variables and Wilcoxon rank-sum/Mann-Whitney U test in detail. **(BT2/CO4) 6**

Q. 5 Solve Any Two of the following. **12**

- A)** A pharmaceutical company conducts an experiment to test the effect of a new cholesterol medication. The company selects 15 subjects randomly from a larger population. In Group 1, subjects receive 0 mg/day; in Group 2, 50 mg/day; and in Group 3, 100 mg/day. After 30 days, doctors measure the cholesterol level of each subject. **(BT3/CO5) 6**
 - i) Does dosage level have a significant effect on cholesterol level?
 - ii) How strong is the effect of dosage level on cholesterol level?

Dosage		
Group 1 0 mg	Group 2 50 mg	Group 3 100 mg
210	210	180
240	240	210
270	240	210
270	270	210
300	270	240

- B)** Explain in detail MANOVA and different test Statistics for MANOVA. **(BT2/CO5) 6**
- C)** What is Non Parametric ANOVA? List the types and explain any one in detail. **(BT2/CO5) 6**

***** Best of Luck *****